

RDT-1B: A DIFFUSION FOUNDATION MODEL FOR BIMANUAL MANIPULATION

2025 ICML

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Introduction

Robot Foundation Model

A large-scale model trained on **massive demonstrations using imitation learning**(supervised learning style), designed to generalize across diverse manipulation tasks.

NLP : Learns words/sentences(tokens of human language), to capture general linguistic knowledge and generalize to unseen language tasks.

CV : Learns pixels/images(tokens of the visual world), to capture general visual representations and generalize to unseen vision tasks.

Robotics : Learns actions/trajectories(tokens of embodied interaction), to capture general physical priors and generalize to unseen manipulation tasks.

RDT-1B:

Architecture

: Diffusion Transformer + Multi-modal Encoding + Robot-specific Modifications + Unified Action Space

Capabilities

: Captures complex multimodal distributions of bimanual actions

: Generalizes across different robots and diverse environments

Goal

: Build a foundation model for robust and generalizable bimanual manipulation

Introduction

Contribution

Largest Diffusion-Based Robotic Model: Pre-trained on multi-robot datasets with 1.2B parameters.

Extensive Bimanual Dataset Fine-Tuning: 6K+ trajectories in a multi-task bimanual dataset enhance manipulation capabilities.

State-of-the-Art Performance: 56% improvement in success rates over strong baselines and excels in challenging tasks.

Exceptional Generalization: Capable of precise control, e.g., manipulating a robot dog with a joystick.

Zero/Few-Shot Learning: Performs well on unseen objects, scenes, instructions, and skills.

→ **Key Success Factors:** Diffusion modeling, Foundation Model(large model size, and large-scale data)

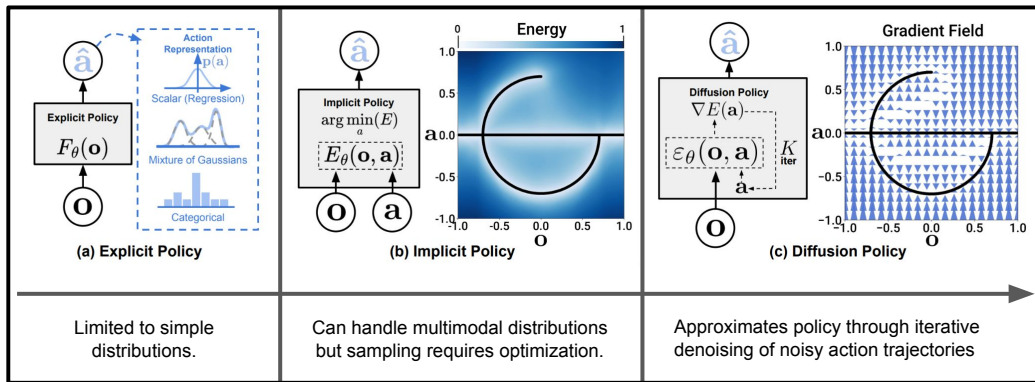
Introduction

Previous Works(Diffusion Policy)

A method that generates robot behavior by modeling visuomotor policies as a conditional denoising diffusion process.

Diffusion Policy: Visuomotor Policy Learning via Action Diffusion, [RSS 2023]

: From direct $\rightarrow$ implicit $\rightarrow$ generative, increasing expressiveness and generalization.



DDPM(Denoising Diffusion Probabilistic Models)

: Forward process

$$q(x_t | x_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t} x_0, (1 - \bar{\alpha}_t) I)$$

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, \quad \epsilon \sim \mathcal{N}(0, I)$$

$$\rightarrow A_t^k = \sqrt{1 - \beta_k} A_t^{k-1} + \sqrt{\beta_k} \epsilon_k$$

: Reverse process

$$\mathbf{x}^{k-1} = \alpha(\mathbf{x}^k - \gamma \epsilon_\theta(\mathbf{x}^k, k) + \mathcal{N}(0, \sigma^2 I))$$

$$\mathbf{x}' = \mathbf{x} - \gamma \nabla E(\mathbf{x})$$

* Interpreted as single noisy gradient descent step

$$\rightarrow \mathbf{A}_t^{k-1} = \alpha(\mathbf{A}_t^k - \gamma \epsilon_\theta(\mathbf{O}_t, \mathbf{A}_t^k, k) + \mathcal{N}(0, \sigma^2 I))$$

* Predicted from network

: Loss function

$$\mathcal{L} = \text{MSE}(\epsilon^k, \epsilon_\theta(\mathbf{x}^0 + \epsilon^k, k))$$

$$\rightarrow \mathcal{L} = \text{MSE}(\epsilon^k, \epsilon_\theta(\mathbf{O}_t, \mathbf{A}_t^0 + \epsilon^k, k))$$

Introduction

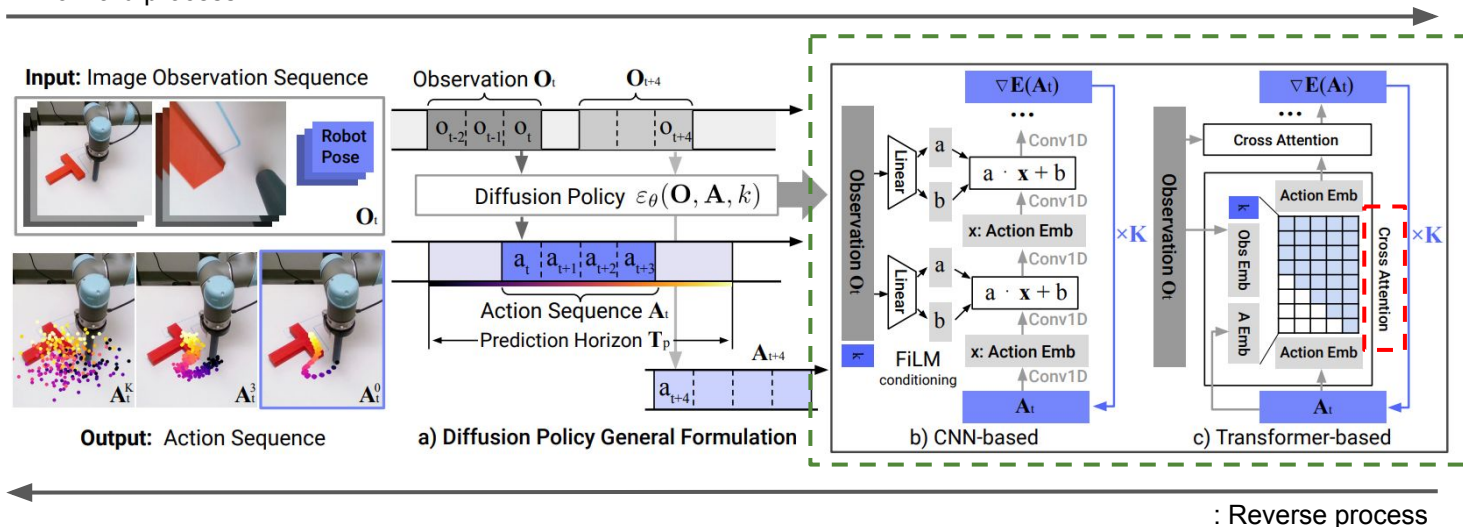
Previous Works(Diffusion Policy)

A method that generates robot behavior by modeling visuomotor policies as a conditional denoising diffusion process.

Diffusion Policy: Visuomotor Policy Learning via Action Diffusion, [RSS 2023]

: A novel approach to generating model behaviors by denoising the robot's fine motor (visual motor) control.

: Forward process



- **CNN**: Stable but struggles with rapidly changing action sequences.
- **Transformer**: Sensitive to sequence dynamics, excels at multimodal distributions, but requires additional hyperparameter tuning.

Related Works

Learning-based Bimanual Manipulation

Challenge in learning a bimanual manipulation policy : **high dimensionality of the action space**, which exacerbates the **data scarcity** (Zollner et al., 2004; Smith et al., 2012; Lioutikov et al., 2016; Stepputtis et al., 2022) the **multi-modal behavior** (Colome´ & Torras, 2018; 2020; Figueroa & Billard, 2017; Sharma et al., 2018; Xie et al., 2020; Franzese et al., 2023)

Attention 1: The more cost-effective interfaces for data collection (Zhao et al., 2023; Aldaco et al., 2024)

Limitation:

- limited to specific hardware configurations
- still insufficient to bridge the data gap for a generalizable policy

Attention 2: To reduce data requirements by introducing inductive biases,

- distinguishing two arms for stabilization and functionality (Grannen et al., 2023b)
- parameterizing movement primitives (Batinica et al., 2017; Amadio et al., 2019; Chitnis et al., 2020; Franzese et al., 2023)
- using voxel representations (Grotz et al., 2024; Liu et al., 2024).

Benefit: use strong priors or simplified modeling, which successfully reduce the action space

Limitation:

- reduced scope of application
- failing to express the multi-modality of bimanual behaviors (Pearce et al., 2023).

Related Works

Foundation Models for Robotics

VLM (Brohan et al., 2022; Driess et al., 2023; Brohan et al., 2023; Collaboration et al., 2023; Kim et al., 2024):

Benefit: generalization to new objects and tasks by action prediction

Cost: In bimanual manipulation,

- quantization errors
- uncoordinated behaviors (Pearce et al., 2023)

Diffusion Model (Ho et al., 2020; Chi et al., 2023; Pearce et al., 2023; Ghosh et al., 2023) :

Benefit : supporting precise and continuous control

Ghosh et al. (2023) pre-train a Transformer-based diffusion policy on a subset of Open X-Embodiment (Collaboration et al., 2023) dataset (25 datasets), with up to 93M parameters.

Problem Statement

Target robot (evaluation platform)

ALOHA dual-arm robot (2 grippers + 3 cameras), applicable to general dual-arm robots

Task definition

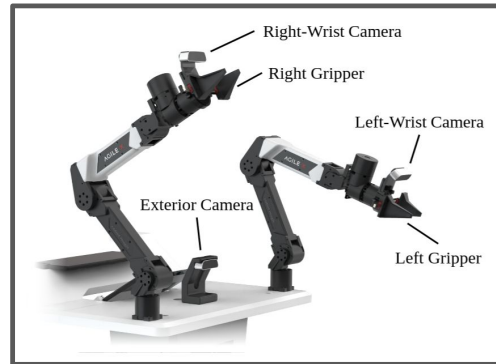
Input: language instruction $\ell \in \mathcal{L}$, observation $o_t := (X_{t-T_{\text{img}}+1:t}, z_t, c)$

Output: action $\pi_\theta : (\ell, o_t) \mapsto a_t, a_t \subseteq z_{t+1}$

Data challenge

Target robot data < **10K trajectories** (too small for foundation model)

Solution: Pre-train on multi-robot dataset $D_{\text{pre}} = \{(\ell^{(i)}, o_t^{(i)}, a_t^{(i)})\}_{i=1}^{N_{\text{pre}}}$, then fine-tune on target $D_{\text{ft}} = \{(\ell^{(j)}, o_t^{(j)}, a_t^{(j)})\}_{j=1}^{N_{\text{ft}}}$



[ALOHA dual-arm robot]

Definition of Notation

Time index : $t \in \mathbb{N}^+$: the current discrete time step

Image observation : X_t : the image observation (or a set of images from multiple cameras) captured at time step t

History length : $T_{\text{img}} \in \mathbb{N}^+$: the fixed number of consecutive image frames included in the observation history

Image history : $X_{t-T_{\text{img}}+1:t} = (X_{t-T_{\text{img}}+1}, X_{t-T_{\text{img}}+2}, \dots, X_t)$: the sequence of size T_{img} RGB images leading up to and including the current frame X_t

Observation tuple: $o_t := (X_{t-T_{\text{img}}+1:t+1}, z_t, c)$ aggregates heterogeneous sensory modalities: vision, proprioception, and control signal

Proprioception : $z_t \in \mathbb{R}^{d_z}$: low-dimensional robot state at time t , (e.g. joint angles, gripper positions, or velocities)

Control frequency : $c \in \mathbb{R}$: control frequency (Hz) at which the robot executes actions

Problem Statement

Proprioceptive Vector of ALOHA MOBILE in paper

$$z_t = [z_t^R, z_t^L, z_t^B, z_t^{\text{resv}}] \in \mathbb{R}^{128}$$

Right Arm Structure :

$$z_t^R = [z_t^{R,\text{arm-pos}}, z_t^{R,\text{grip-pos}}, z_t^{R,\text{arm-vel}}, z_t^{R,\text{grip-vel}}, z_t^{R,\text{eef-pos}}, z_t^{R,\text{eef-6D}}, z_t^{R,\text{eef-vel}}, z_t^{R,\text{eef-angvel}}, z_t^{R,\text{resv}}]$$

Left Arm Structure :

$$z_t^L = [z_t^{L,\text{arm-pos}}, z_t^{L,\text{grip-pos}}, z_t^{L,\text{arm-vel}}, z_t^{L,\text{grip-vel}}, z_t^{L,\text{eef-pos}}, z_t^{L,\text{eef-6D}}, z_t^{L,\text{eef-vel}}, z_t^{L,\text{eef-angvel}}, z_t^{L,\text{resv}}]$$

Base Structure :

$$z_t^B = [z_t^{B,\text{lin-vel}}, z_t^{B,\text{ang-vel}}] \begin{cases} z_t^{B,\text{lin-vel}} \in \mathbb{R}^2 \text{ (linear velocity on the plane)} \\ z_t^{B,\text{ang-vel}} \in \mathbb{R}^1 \text{ (yaw angular velocity)} \end{cases}$$

Dimension Breakdown : 50 (Right) + 50 (Left) + 3 (Base) + 25 (Reserved) = 128

Action Space a_t :

The action at time t is defined as a subset of the next proprioception state: $a_t \subseteq z_{t+1}$

Index Range	Element Index	Mapped Physical Quantity
[0, 10)	0–9	Right arm joint positions
[10, 15)	10–14	Right gripper joint positions
[15, 25)	15–24	Right arm joint velocities
[25, 30)	25–29	Right gripper joint velocities
[30, 33)	30–32	Right end effector positions
[33, 39)	33–38	Right end effector 6D pose
[39, 42)	39–41	Right end effector velocities
[42, 45)	42–44	Right end effector angular velocities
[45, 50)	45–49	Reserved
[50, 60)	50–59	Left arm joint positions
[60, 65)	60–64	Left gripper joint positions
[65, 75)	65–74	Left arm joint velocities
[75, 80)	75–79	Left gripper joint velocities
[80, 83)	80–82	Left end effector positions
[83, 89)	83–88	Left end effector 6D pose
[89, 92)	89–91	Left end effector velocities
[92, 95)	92–94	Left end effector angular velocities
[95, 100)	95–99	Reserved
[100, 102)	100–101	Base linear velocities
[102, 103)	102	Base angular velocities
[103, 128)	103–127	Reserved

Challenge of robot foundation model

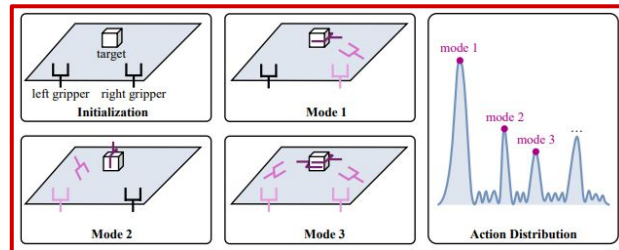
Challenges 1 : Architecture Design

Expressiveness

- Bimanual actions have **multi-modal distributions** (many possible modes to complete the same task).
- Simple regression → leads to **invalid “average” actions**.
- Need a model that can **capture complex, multi-modal action distributions**.

Scalability

- Must handle **multi-modal inputs** (language, vision, actions)
- Requires **stable large-scale training** on heterogeneous datasets



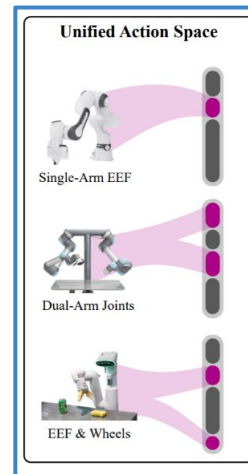
Challenge 2: Training on Heterogeneous Data

Problem

- Robots differ in **physical structures** and **action spaces**.
- Naïve solutions: use only similar robots or shared features → **lose diversity** and harm generalization.

Need

- A **Physically Interpretable Unified Action Space** to align diverse robot actions.
- Enables the model to learn **transferable physical knowledge** across robots



Training Dataset

[Pre-training datasets]

Pre-Training Dataset	Sample Percentage (%)
RT-1 Dataset (Brohan et al., 2022)	9.00
TACO Dataset (Rosete-Beas et al., 2022)	1.99
JACO Play Dataset (Dass et al., 2023)	1.10
Cable Routing Dataset (Luo et al., 2023)	0.27
NYU Door Opening (Pari et al., 2021)	0.33
Viola (Zhu et al., 2022a)	0.40
Berkeley UR5 (Chen et al.)	1.06
TOTO (Zhou et al., 2023)	1.06
Kuka (Kalashnikov et al., 2018)	1.66
Language Table (Lynch et al., 2022)	3.32
Columbia Cairlab Pusht Real (Chi et al., 2023)	0.40
Stanford Kuka Multimodal Dataset (Lee et al., 2019)	1.83
Stanford Hydra Dataset (Belkale et al., 2023)	0.80
Austin Buds Dataset (Zhu et al., 2022b)	0.23
Maniskill Dataset (Gu et al., 2023)	5.78
Furniture Bench Dataset (Heo et al., 2023)	2.36
UCSD Kitchen Dataset (Yan et al., 2023)	0.40
UCSD Pick And Place Dataset (Feng et al., 2023)	1.23
Austin Sailor Dataset (Nasiriany et al., 2022)	0.50
Austin Sirius Dataset (Liu et al., 2023)	0.80
BC Z (Jang et al., 2021)	6.91
UTokyo PR2 Opening Fridge (Oh et al., 2023)	0.30
UTokyo PR2 Tabletop Manipulation (Oh et al., 2023)	0.50
UTokyo Xarm Pick And Place (Matsushima et al., 2023)	0.33
UTokyo Xarm Bimanual (Matsushima et al., 2023)	0.03
Berkeley MVP (Radosavovic et al., 2022)	0.73
Berkeley RPT (Radosavovic et al., 2022)	1.00
KAIST Nonprehensile (Kim et al., 2023)	0.46
Tokyo U LSMO (Osa, 2022)	0.23
DLR Sara Grid Clamp (Padalkar et al., 2023)	0.03
Robocook (Shi et al., 2023)	1.66
ImperialCollege Sawyer Wrist Cam (RethinkRobotics)	0.43
Iamlab CMU Pickup Insert (Saxena et al., 2023)	0.83
UTAustin Mutex (Shah et al., 2023b)	1.29
Fanuc Manipulation (Zhu et al., 2023)	0.66
Play Fusion (Chen et al., 2023)	0.80
DROID (Khazatsky et al., 2024)	10.06
FMB (Luo et al., 2024)	1.39
Dobb-E (Shafiullah et al., 2023)	1.20
QUT Dexterous Manipulation (Federico Ceola, 2023)	0.46
Aloha Dataset (Zhao et al., 2023)	4.98
Mobile Aloha Dataset (Fu et al., 2024)	4.98
RoboSet (Kumar et al., 2024)	4.48
RH20T (Fang et al., 2023)	10.99
Calvin Dataset (Mees et al., 2022)	3.32
BridgeData V2 (Walke et al., 2023)	7.44

[Fining Tuning datasets]

Overview



Bimanual



6-DoF Joint
Pos. & Vel.



3 Camera
Views



300+ Tasks



6K+ Episodes



3M+ Frames

Diverse Objects & Scenes

100+ Diverse Objects



15+ Diverse Scenes



Various Lighting



Challenging Tasks

Bimanual



Open the drawer



Get cold water from the dispenser



Interlock the blue slippers



Pull out chips from the green bucket

Comprehension



Pick the largest numbered chip



Spell "love" with the letters



Solve the equation



Put the doll in least similar size into box

Dexterity



Control robot dog to walk straight



Open the zipper of the file bag



Plug the charging cable into the phone



Unscrew the cap from plastic bottle

Augmented Instructions



Human Annotation:
Pick up the ice scoop on the desk filled with ice cubes, pour the ice into the goblet, and finally put the scoop back on the table.



Expanded Annotation:
Carefully grasp the ice scoop resting on the desk, which is filled with ice cubes, gently transfer the ice cubes into the goblet without spilling, and then precisely place the scoop back in its original position on the desk.

+

Simplified Annotation:
Pour ice cubes from the ice scoop into the goblet.

Pre-training dataset:

46+ public robot datasets (1M+ trajectories, 21TB) across diverse robots, tasks, and modalities for transferable priors

Fine-tuning dataset:

6K+ trajectories (300+ tasks, 100+ objects, 15+ scenes) on the ALOHA dual-arm with augmented instructions

Robotics Diffusion Transformer - diffusion model

Problem Statement

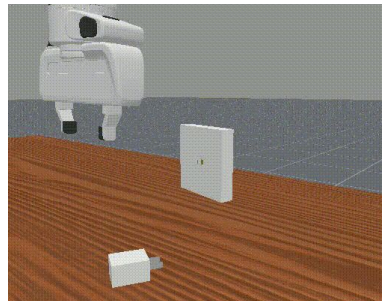
Traditional Method (Regressive Method)

$$\text{Model Learning : } (\ell, o_t) \rightarrow a_t \longrightarrow \min \mathbb{E} \left[\|a_t - f(\ell, o_t)\|_2^2 \right] \longrightarrow f(\ell, o_t) = \mathbb{E}[a_t | \ell, o_t]$$

Training Dataset : (ℓ, o_t, a_t)

* Deterministic infeasible actions

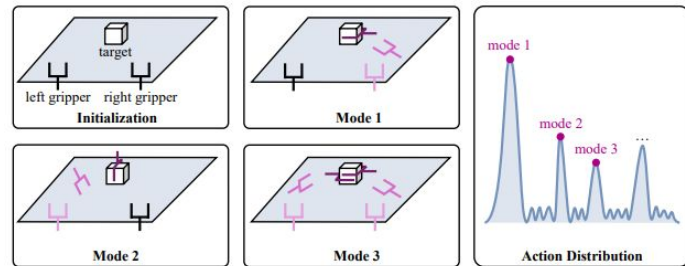
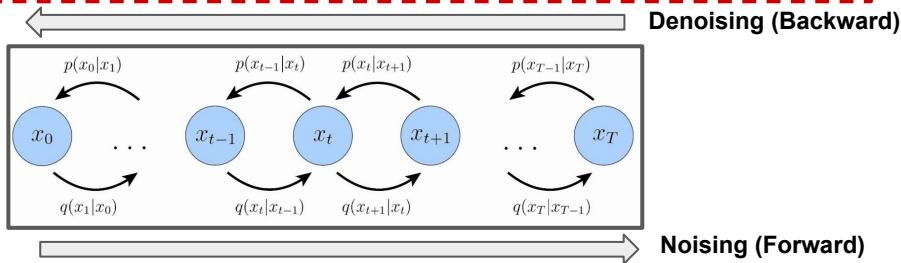
Deterministic regression fails in OOD by averaging modes into infeasible actions



Paper Method (Diffusion Method)

Model Learning : $a_t \sim p(a_t | \ell, o_t) \longrightarrow$ remain the possible way

* Learn total probabilistic distribution



Initialize : $a_t^k \sim \mathcal{N}(0, I), \beta_k := 1 - \alpha_k, \alpha_{k-1} := \prod_{i=1}^{k-1} \alpha_i, z \sim \mathcal{N}(0, I)$ **Loss function :** $\mathcal{L}(\theta) = \mathbb{E}_{(\ell, o_t, a_t), k, \epsilon} [\|a_t - f_\theta(\ell, o_t, \tilde{a}_t, k)\|_2^2]$

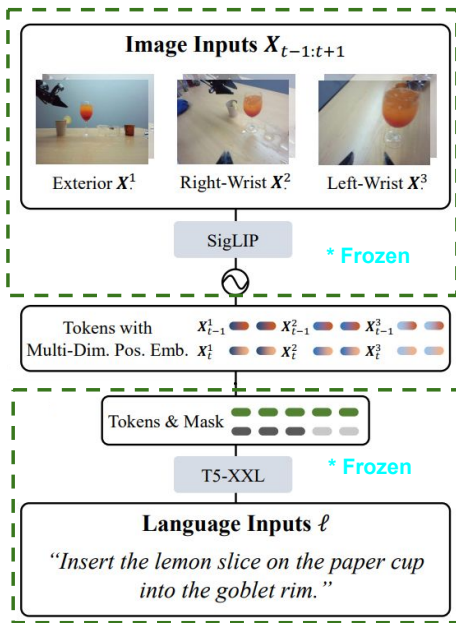
Backward Process : $a_t^{k-1} = \sqrt{\frac{\bar{\alpha}_{k-1}\beta_k}{1-\bar{\alpha}_k}} a_t^0 + \sqrt{\frac{\alpha_k(1-\bar{\alpha}_{k-1})}{1-\bar{\alpha}_k}} a_t^k + \sigma_k z, k=K, \dots, 1$ $a_t^0 \approx f_\theta(\ell, o_t, a_t^k, k)$

* Clean action at a_t^0 is only available after full sampling $\rightarrow$ intractable during intermediate steps. $\rightarrow$ run using neural network

Robotics Diffusion Transformer - Architecture Details

Encoding of Heterogeneous Multi-Model Inputs

Paper Method (Diffusion Method)



Problem: Modality **heterogeneity** (format & dimensionality) $\rightarrow$ harder **multimodal training**.

Strategy: Map each modality with a dedicated encoder into a **unified latent space**.

Low-Dim Inputs: proprioception / action chunk / control frequency
 $\rightarrow$ **MLP + Fourier features** (capture high-frequency signals).

Image Inputs: high-dimensional and contain rich spatial and semantic information
SigLIP (image-text aligned) encoder, **frozen weights** (save GPU memory).

Language Inputs: T5-XXL encoder, **frozen weights** (save GPU memory).

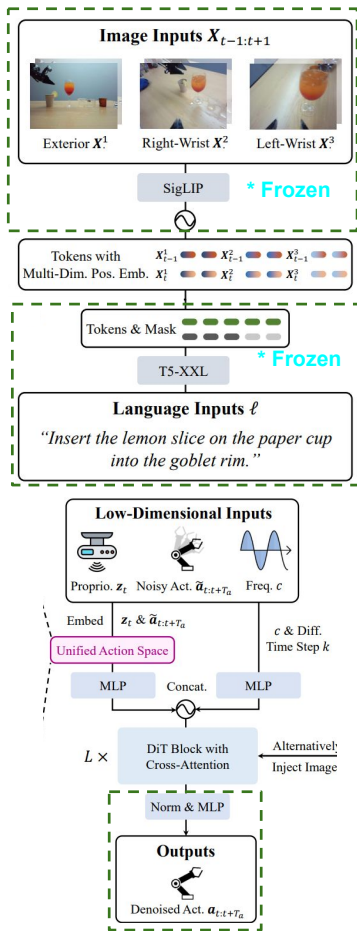
Heterogeneity-1 (cross-modal):

Modality capacity imbalance; vision $\gg$ text; token count skew; risk: over-reliance on vision.

Heterogeneity-2 (intra-modal):

View-level variance; exterior = wide FOV/global context vs wrist = narrow FOV/fine detail;
risk: shortcut on exterior, neglect wrist.

ROBOTICS DIFFUSION TRANSFORMER



Modality	Encoder	Trainable	Adaptor
Language	T5-XXL (Raffel et al., 2020)	N	2-layers MLP
Image	SigLIP (Zhai et al., 2023)	N	2-layers MLP
Action	-	-	3-layers MLP

Use short histories for efficiency: image history $T_{\text{img}} = 2$, action chunk $T_a = 64$

Token space sizes are heterogeneous: language 4096, image 1152, RDT 2048

Apply **adaptors** with **GeLU** to project each modality to the shared RDT token dimension 2048 for clean fusion/cross-attention.

$$\text{Adaptor}_{\text{lang}}: \mathbb{R}^{4096} \rightarrow \mathbb{R}^{2048}, \text{Adaptor}_{\text{img}}: \mathbb{R}^{1152} \rightarrow \mathbb{R}^{2048}, \phi = \text{GeLU}$$

Action Chunking for Stable Long-Horizon Control

Fewer decisions → less error accumulation

Joint prediction → better temporal consistency, prevents abrupt action jumps

Low-dimensional actions → diffusion over chunks adds little latency

Solution : Predict an action chunk of T_a steps, $\mathbf{A}_{t:t+T_a-1}$ at once to reduce the number of decision steps

$$\hat{\mathbf{A}}_{t:t+T_a-1} = f_{\theta}(\ell, o_t, \tilde{\mathbf{A}}_{t:t+T_a-1}, k) \quad \mathcal{L}(\theta) = \mathbb{E} \left[\frac{1}{T_a} \sum_{\tau=0}^{T_a-1} \|a_{t+\tau} - \hat{a}_{t+\tau}\|_2^2 \right], \quad \hat{a}_{t+\tau} = [f_{\theta}(\cdot)]_{\tau}$$

Robotics Diffusion Transformer - Architecture Details

Low-Dimensional (Unified) Embeddings

Embed proprioception, noisy chunk, and scalar contexts into a shared token width d

$$u_t^z = \text{MLP}_z(z_t) \in \mathbb{R}^d, \quad u_t^a = \text{MLP}_a(\tilde{A}_{t:t+T_a}) \in \mathbb{R}^{T_a \times d}, \quad u^c = \text{MLP}_c(c) \in \mathbb{R}^d, \quad u^k = \text{MLP}_k(k) \in \mathbb{R}^d$$

Concatenate along length (in-context conditioning), then add positional encoding

$$U_t = [u_t^z \parallel u_t^a \parallel u^c \parallel u^k] \in \mathbb{R}^{(1+T_a+1+1) \times d}, \quad \tilde{U}_t = U_t + \text{PosEnc}(U_t)$$

Image Tokens with Grid Positional Encoding

For time index $i \in \{1, \dots, T_{\text{img}}\}$, camera $j \in \{1, \dots, N_{\text{cam}}\}$, patch $p \in \{1, \dots, N_{\text{patch}}\}$
 $v_{i,j,p} = \text{SigLIP}(X_i^{(j)})[\text{patch } p]$, $\hat{v}_{i,j,p} = \text{MLP}_{\text{img}}(v_{i,j,p}) \in \mathbb{R}^d$, $\tilde{v}_{i,j,p} = \hat{v}_{i,j,p} + \Phi_{\text{grid}}(i, j, p) \in \mathbb{R}^d$

Stack to form $\tilde{V} \in \mathbb{R}^{(T_{\text{img}} N_{\text{cam}} N_{\text{patch}}) \times d}$

Language Token

$$L = \text{T5-XXL}(\ell) \in \mathbb{R}^{L_\ell \times d_\ell}, \quad \hat{L} = \text{MLP}_{\text{lang}}(L) \in \mathbb{R}^{L_\ell \times d}, \quad \tilde{L} = \text{Mask}(\hat{L}; p_{\text{drop}} = 0.1)$$

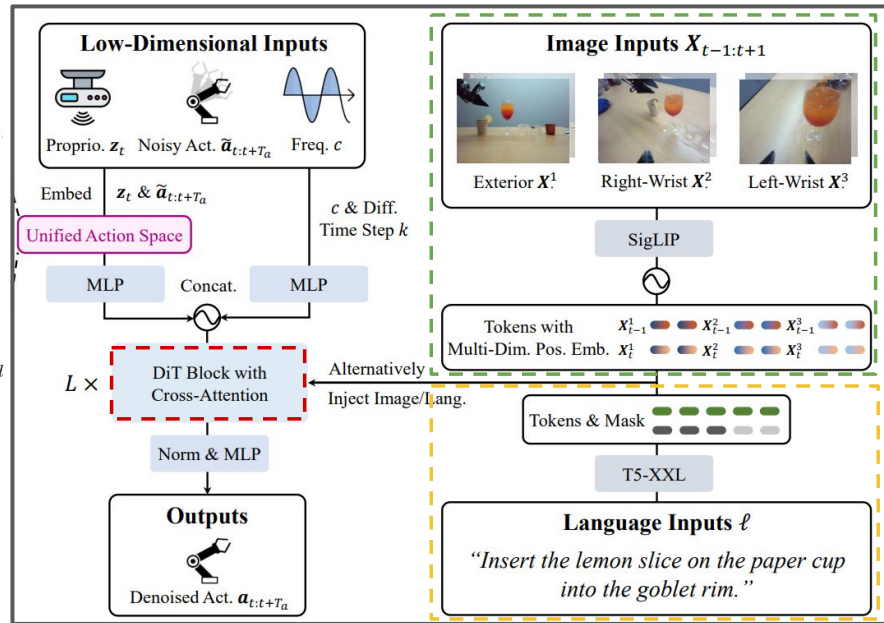
Apply a padding mask in attention to hide PAD token

Token Assembly and DiT Backbone

Assemble all token : $\mathcal{T}_0 = [\tilde{U}_t \parallel \tilde{V} \parallel \tilde{L}] \in \mathbb{R}^{N_{\text{tok}} \times d}$, $N_{\text{tok}} = (1 + T_a + 1 + 1) + T_{\text{img}} N_{\text{cam}} N_{\text{patch}} + L_\ell$

Process through DiT blocks with RMSNorm, QKNorm, and alternating cross-attention:

$$\begin{aligned} H' &= \text{SelfAttn}^{\text{QKNorm}}(\text{RMSNorm}(H)) + H, \\ H'' &= \begin{cases} \text{CrossAttn}^{\text{QKNorm}}(\text{RMSNorm}(H'), \tilde{V}) + H', & \text{odd block, } H \leftarrow H^+, H_0 = \mathcal{T}_0, H_L \text{ after } L \text{ blocks} \\ \text{CrossAttn}^{\text{QKNorm}}(\text{RMSNorm}(H'), \tilde{L}) + H', & \text{even block,} \end{cases} \\ H^+ &= \text{MLP}(\text{RMSNorm}(H'')) + H'', \end{aligned}$$



Robotics Diffusion Transformer - Architecture Details

Output Head and Final Mapping

Select the low-dimensional slice of HL and decode to actions :

$$\hat{A}_{t:t+T_a} = \text{MLP}_{\text{dec}}\left(\text{RMSNorm}(\text{Slice}_{\text{low}}(H_L))\right) \in \mathbb{R}^{T_a \times d_a}$$

Summary

Select the low-dimensional slice of HL and decode to actions :

$$\hat{A}_{t:t+T_a} = f_{\theta}(\ell, X_{t-T_{\text{img}}+1:t}, z_t, c, k, \tilde{A}_{t:t+T_a})$$

Training-Time Modality Dropout

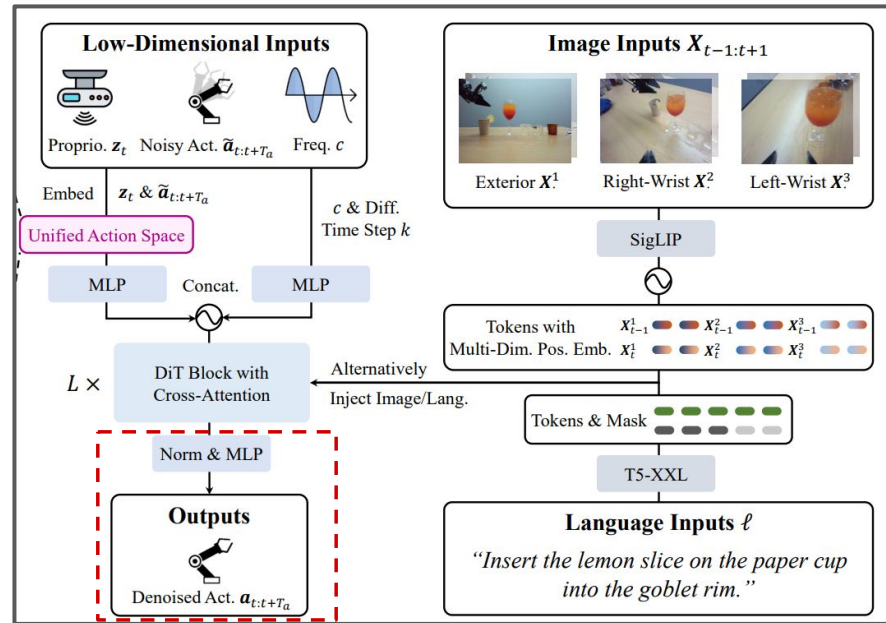
Independently drop (mask) each modality with probability 0.1 :

$$\tilde{U}_t \leftarrow \text{Mask}(\tilde{U}_t), \quad \tilde{V} \leftarrow \text{Mask}(\tilde{V}), \quad \tilde{L} \leftarrow \text{Mask}(\tilde{L})$$

Shape Summary

$$U_t \in \mathbb{R}^{(1+T_a+1+1) \times d}, \quad \tilde{V} \in \mathbb{R}^{(T_{\text{img}} N_{\text{cam}} N_{\text{patch}}) \times d}, \quad \tilde{L} \in \mathbb{R}^{L_{\ell} \times d},$$

$$\mathcal{T}_0, H_L \in \mathbb{R}^{N_{\text{tok}} \times d}, \quad \hat{A}_{t:t+T_a} \in \mathbb{R}^{T_a \times d_a}$$



Action tokens : U_t , Image Patches : $\tilde{V}$, Language Token : $\tilde{L}$, Learned Embeddings : $\mathcal{T}_0, H_L$, Predicted Action Sequence : $\hat{A}_{t:t+T_a}$

Robotics Diffusion Transformer - Architecture Details

Encoding of Heterogeneous Multi-Model Inputs

QKNorm & RMSNorm (Stability):

Robot states/actions have unstable scales $\rightarrow$ attention/gradient blow-ups.

QKNorm: normalize q,k,q,k before dot-product to stabilize attention : $\hat{Q} = \frac{Q}{\text{RMS}(Q)}$, $\hat{K} = \frac{K}{\text{RMS}(K)}$, $\text{Attn} = \text{softmax}\left(\frac{\hat{Q}\hat{K}^\top}{\sqrt{d_k}}\right)V$.

RMSNorm: no centering $\rightarrow$ preserves time-series symmetry, avoids token/attention shift : $\text{RMSNorm}(x) = \gamma \cdot \frac{x}{\sqrt{\frac{1}{d} \sum_{i=1}^d x_i^2 + \epsilon}}$, $x \in \mathbb{R}^d$, $\gamma \in \mathbb{R}^d$

MLP Decoder (Nonlinear Projection): $\hat{a}_{t:t+H} = \text{MLP}(h_{t:t+H})$, $\text{MLP} = \sigma(W_2 \sigma(W_1 \cdot + b_1) + b_2)$

Conditions are high-dim & variable-length (image $\gg$ text); single-token compression loses info.

Use **cross-attention** for full-length conditions, and **alternate** image/text per layer to avoid text overshadowing.

Boosts instruction-following.

Alternating Condition Injection (ACI):

Conditions are high-dim & variable-length (image $\gg$ text); single-token compression loses info.

Use **cross-attention** for full-length conditions, and **alternate** image/text per layer to avoid text overshadowing.

Boosts instruction-following.

$$x^{(0)} = x, \quad x^{(i)} = \text{Block}_i(x^{(i-1)}, c^{(i)}), \quad i = 1, \dots, L$$

$$c^{(i)} = \begin{cases} c_{\text{text}}, & i = \text{even} \\ c_{\text{img}}, & i = \text{odd} \end{cases} \quad \text{Block}_i: \text{SelfAttn} \rightarrow \text{CrossAttn}(c^{(i)}) \rightarrow \text{FFN}$$

* **notation** B : batch, D : hidden size, H : horizon, $c_{\text{text}} \in \mathbb{R}^{B \times L_{\text{text}} \times D}$, $c_{\text{img}} \in \mathbb{R}^{B \times L_{\text{img}} \times D}$

Robotics Diffusion Transformer - Training Detail

Unified Action Space with Padding Indicator

embed each robot's action into a 128-D unified space and avoid semantic collision of zero (e.g., true velocity = 0 vs. padded 0) by concatenating availability masks

Per-modality padded vectors and mask

$$a_t^{\text{pad}} \in \mathbb{R}^{128}, \quad z_t^{\text{pad}} \in \mathbb{R}^{128}, \quad m^a, m^z \in \{0, 1\}^{128}$$

where $m_i = 1$ denotes a valid dimension and $m_i = 0$ a padded one

Token encoder input

$$\tilde{a}_t = [a_t^{\text{pad}}; m^a] \in \mathbb{R}^{256}, \quad \tilde{z}_t = [z_t^{\text{pad}}; m^z] \in \mathbb{R}^{256}$$

These are then projected into the model token space

Training Inspection and Early Stopping

At regular intervals, run diffusion sampling and compare sampled actions to ground truth

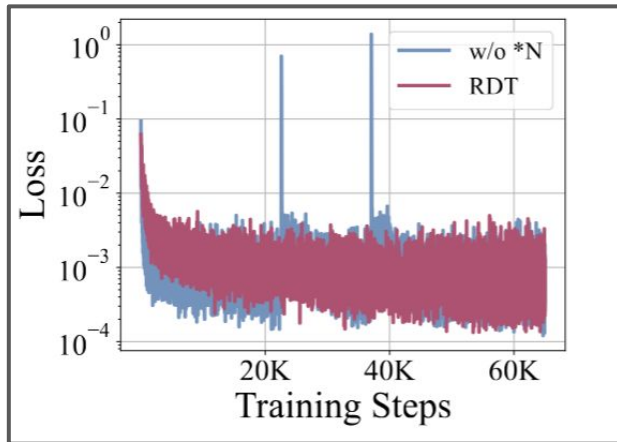
Monitoring MSE

For a validation set $\mathcal{V}$, $\hat{A}_{t:t+T_a} \sim p_\theta(\cdot \mid \ell, o_t)$,

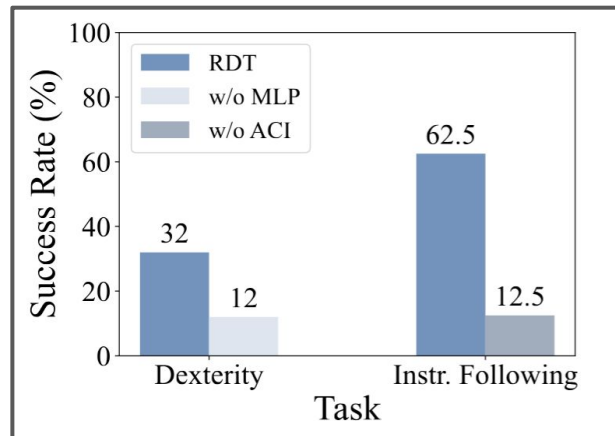
$$\text{MSE}(\theta) = \frac{1}{|\mathcal{V}|} \sum_{(\ell, o, A) \in \mathcal{V}} \frac{\|A_{t:t+T_a} - \hat{A}_{t:t+T_a}\|_2^2}{T_a}$$

Empirically, MSE shows a positive correlation with deployment performance

Ablation Study



[Loss w/o QKN & RMSN]



[Task w/o MLP or ACI]

Loss w/o QKN & RMSN

shows unstable loss **without QKNorm & RMSNorm**, thus QKNorm & RMSNorm fixes exploding/oscillating loss.

Task w/o MLP or ACI

Tasks: **Robot Dog (walk straight)**, **Pour Water-L-1/3 (correct amount)**.

Models **without pre-training** (resource-constrained ablation)

w/o MLP Decoder ↓: linear head underfits **nonlinear robot dynamics** → lower success, especially in precise/dexterous control.

w/o ACI ↓: injecting image+text together every layer → **image tokens overshadow text** → weaker instruction following.

Full RDT (MLP Decoder + **Alternating Condition Injection**) performs best.

Ablation Study

Research Question

1) Zero-Shot Generalization

- Can RDT perform well on **unseen objects and scenes**, even without specific training data?

2) Instruction Following

- How effectively can RDT **understand and execute novel instructions** across **unseen modalities**?

3) Few-Shot Learning

- Can RDT **quickly acquire new skills** with only **1–5 demonstrations**, unlike conventional models?

4) Dexterous Manipulation

- Is RDT capable of **fine-grained and delicate operations**, such as controlling tools or handling deformable objects?

5) Scaling Effects

- Do **large model sizes, massive datasets, and diffusion-based modeling** significantly improve RDT's performance compared to smaller or regression-based variants?

TASK NAME	DIMENSION	EXPLANATION
Wash Cup	Unseen Object (<i>Q1</i>)	To wash one seen and two unseen cups with the faucet
Pour Water	Unseen Scene (<i>Q1</i>)	To pour water into the cup in three unseen rooms
Pour Water-L-1/3	Instruction Following (<i>Q2</i>)	To pour water into the cup with the left hand until one-third full
Pour Water-R-2/3	Instruction Following (<i>Q2</i>)	To pour water into the cup with the right hand until two-thirds full
Handover	5-Shot Learning (<i>Q3</i>)	To move the marker to the box, where handover is needed due to far distance
Fold Shorts	1-Shot Learning (<i>Q3</i>)	To fold the shorts in half horizontally
Robot Dog	Dexterity (<i>Q4</i>)	To push the joystick straight to control the robot dog to walk in a straight line

[Dimensions when designing tasks]

Experiment

Ablation Study for Q5: Examine how **model size**, **pre-training**, and **diffusion modeling** contribute to RDT’s performance.

METHOD NAME	LARGE MODEL	LARGE MULTI-ROBOT DATA	MODELING
RDT (scratch)	✓	✗	Diffusion
RDT (small)	✗	✓	Diffusion
RDT (regress)	✓	✓	Regression
RDT (ours)	✓	✓	Diffusion

[Comparison of different baselines]

VARIANT NAME	UNSEEN OBJECT	UNSEEN SCENE	INSTRUCTION FOLLOWING
RDT (regress)	12.5	50	12.5
RDT (small)	37.5	62.5	25
RDT (scratch)	0	25	62.5
RDT (ours)	50	62.5	100

[Ablation study results]

Experimental Setup (Variants)

- RDT (ours):** Full model (1.2B parameters, diffusion-based, pre-trained)
- RDT (small):** Reduced capacity model with 166M parameters
- RDT (regress):** Diffusion removed → deterministic mapping from inputs to actions
- RDT (scratch):** Trained from scratch without pre-training

Evaluation Dimensions

Unseen Object: Wash Cup (unseen cup), **Unseen Scene:** Pour Water (unseen room), **Instruction Following:** Pour Water-L-1/3 (correct amount)

Evaluation Dimensions

All three factors, large model size, large-scale multi-robot data, and diffusion modeling are essential for RDT’s strong performance. Ablation results show sharp performance drops whenever one factor is removed. Pre-training is especially critical: without it (RDT-scratch), the model fails to generalize to unseen objects and scenes.

Experiment

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Experiment



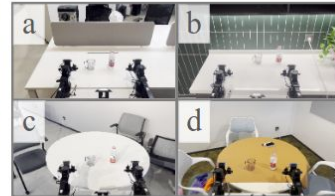
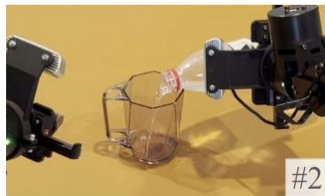
Wash Cup: “Fill the mug with water from the faucet and pour it into the sink.” The mug is randomized up to 10cm. The robot needs to Pick Up Cup (#1), Turn On Faucet (#2), Get Water (#3, to ensure that the water falls into the cup), Pour Out Water (#4), and Place Back Up (#5). The last image shows seen and unseen cups (from left to right).

Wash Cup: seen cup 1 unseen cup 1 unseen cup 2 (Unseen Object)																		
	Pick Up Cup			Turn On Faucet			Get Water			Pour Out Water			Place Back Cup			Total		
ACT	50	12.5	37.5	0	0	0	0	0	0	0	0	0	37.5	0	0	0	0	0
OpenVLA	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Octo	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
RDT (scratch)	37.5	12.5	0	0	12.5	12.5	0	0	0	37.5	12.5	0	25	0	0	0	0	0
RDT (ours)	87.5	87.5	50	62.5	75	50	50	75	50	87.5	75	50	87.5	62.5	50	50	75	50

Q1: Can RDT zero-shot generalize to unseen objects and scenes

RDT shows strong zero-shot generalization to unseen objects/scenes, maintaining high success on Wash Cup & Pour Water unseen settings with a small seen–unseen gap, while baselines often fail to complete the tasks.

Experiment



Pour Water: “Pour water from the bottle into the mug.” The bottle and mug are randomized up to 6cm. The robot needs to Pick Up Bottle (#1), Pour Water (#2), and Place Back Bottle (#3). The last image shows one seen room (a) and three unseen rooms (b-d).

	Pour Water: unseen room 1 unseen room 2 unseen room 3 (Unseen Scene)											
	Pick Up Bottle			Pour Water			Place Back Bottle			Total		
ACT	25	87.5	25	0	50	12.5	0	37.5	12.5	0	37.5	12.5
OpenVLA	0	0	0	0	0	0	0	0	0	0	0	0
Octo	50	0	12.5	12.5	0	0	12.5	0	0	12.5	0	0
RDT (scratch)	62.5	100	62.5	25	87.5	37.5	25	75	25	25	75	25
RDT (ours)	62.5	100	62.5	62.5	100	62.5	62.5	100	62.5	62.5	100	62.5

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Experiment



Pour Water-L-1/3 (-R-2/3): “Pour water from the bottle into the mug until about one-third (two-thirds) with the left (right) hand.” In the left-hand task, the bottle is initially placed to the left of the mug, and vice versa. The bottle and mug are randomized up to 6cm.

Pour Water-L-1/3 | Pour Water-R-2/3 (Instruction Following)

	Pick Up Bottle		Pour Water		Place Back Bottle		Total		Correct Hand		Correct Amount	
OpenVLA	50	0	0	0	0	0	0	0	50	0	0	0
Octo	0	0	0	0	0	0	0	0	0	0	0	0
RDT (scratch)	100	75	75	25	62.5	25	62.5	25	100	75	62.5	12.5
RDT (ours)	100	87.5	100	87.5	100	87.5	100	87.5	100	87.5	100	75

Q2: Can RDT zero-shot generalize to unseen modalities / instructions (e.g., hand-specific, numeric)?

RDT zero-shots to unseen instruction modalities, precisely following hand-specific and numeric commands (e.g., 1/3, 2/3) despite never seeing those phrases by large-scale pretraining, ACI, and expressive diffusion policy.

Experiment



Handover: “Pick up the black marker on the right and put it into the packaging box on the left.” The marker pen and box are randomized up to 3cm. The robot needs to Pick Up Pen (#1), Switch Hand (#2), Drop Pen (#3), and ensure it can Fall into Box (#4).



Fold Shorts: “Fold the basketball shorts into a rectangle.” The shorts are randomized up to 3cm and wrinkles are also randomized.

	Handover (5-Shot)					Robot Dog (Dexterity)				Fold Shorts (1-Shot)
	Pick Up Pen	Switch Hand	Drop Pen	Fall into Box	Total	Grab Remote	Push Joystick	Total	Walk Straight	Total
ACT	44	0	0	0	0	88	32	32	32	0
OpenVLA	0	0	0	0	0	84	0	0	0	0
Octo	12	0	0	0	0	100	4	4	0	4
RDT (scratch)	88	32	24	16	16	100	64	64	32	40
RDT (ours)	100	56	56	40	40	100	76	76	48	68

Q3: Can RDT learn new skills with few shots?

RDT zero-shots to unseen instruction modalities, precisely following hand-specific and numeric commands (e.g., 1/3, 2/3) despite never seeing those phrases by large-scale pretraining, ACI, and expressive diffusion policy.

Experiment



Robot Dog: “Push the left joystick forward to make the robot dog walk straight forward.” The remote control is randomized up to 4cm and the robot dog is up to 50cm. The robot needs to Grab Remote (#1) and Push Joystick (#2) as straight as possible to make the robot dog walk in a straight line as shown in the last two images.

	Handover (5-Shot)					Robot Dog (Dexterity)				Fold Shorts (1-Shot)
	Pick Up Pen	Switch Hand	Drop Pen	Fall into Box	Total	Grab Remote	Push Joystick	Total	Walk Straight	Total
ACT	44	0	0	0	0	88	32	32	32	0
OpenVLA	0	0	0	0	0	84	0	0	0	0
Octo	12	0	0	0	0	100	4	4	0	4
RDT (scratch)	88	32	24	16	16	100	64	64	32	40
RDT (ours)	100	56	56	40	40	100	76	76	48	68

Q4: Can RDT handle dexterous tasks?

RDT achieves more stable, precise dexterous control than baselines by modeling multi-modal, nonlinear actions with a diffusion-based architecture and leveraging large-scale pretraining for robust vision–language grounding (e.g., correctly identifying the black joystick), reducing deviation in tasks like straight walking.

감사합니다.